(ii) (a) Slope of $BC = \frac{1+2}{5-6} = \frac{3}{-1} = -3$

(b) Slope of $AD = \frac{1+2}{2-1} = \frac{3}{1} = 3$

(iii) The slope of AB and CD are equal so AB, CD are parallel.

Similarly the lines AD and BC are not parallel, since their slopes are not equal.

So, we can deduce that the quadrilateral ABCD is a trapezium.

Example 5.14 Consider the graph representing growth of population (in crores). Find the slope of the line AB and hence estimate the population in the year 2030?

Solution The points $A(2005, 96)$ and $B(2015, 100)$ are on the line AB.

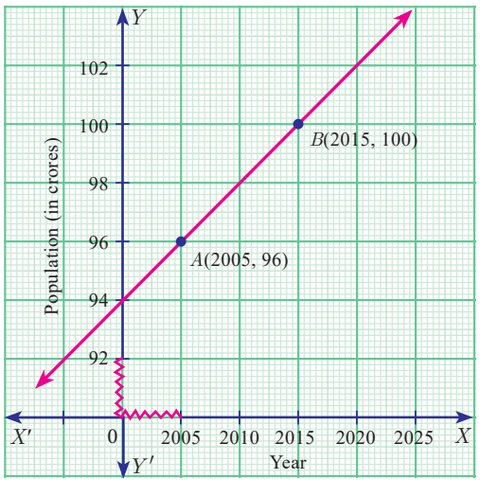


Fig. 5.24

$$\text{Slope of AB} = \frac{100-96}{2015-2005} = \frac{4}{10} = \frac{2}{5}$$

Let the growth of population in 2030 be k crores.

Assuming that the point $C(2030, k)$ is on AB,

we have, slope of AC = slope of AB

$$\frac{k-96}{2030-2005} = \frac{2}{5} \quad \text{gives} \quad \frac{k-96}{25} = \frac{2}{5}$$

$$k - 96 = 10$$

$$k = 106$$

Hence the estimated population in 2030 = 106 Crores.

Example 5.15 Without using Pythagoras theorem, show that the points $(1, -4)$, $(2, -3)$ and $(4, -7)$ form a right angled triangle.

Solution Let the given points be $A(1, -4)$, $B(2, -3)$ and $C(4, -7)$.

$$\text{The slope of AB} = \frac{-3+4}{2-1} = \frac{1}{1} = 1$$

$$\text{The slope of BC} = \frac{-7+3}{4-2} = \frac{-4}{2} = -2$$

$$\text{The slope of AC} = \frac{-7+4}{4-1} = \frac{-3}{3} = -1$$

Slope of $AB \times$ slope of $AC = (1)(-1) = -1$

AB is perpendicular to AC. $\angle A = 90°$

Therefore, ΔABC is a right angled triangle.

Thinking Corner

Provide three examples of using the concept of slope in real-life situations.

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